University of Minho
School of Engineering



Data Security

Master's in Cybersecurity

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Data Security

Objectives

- Understand how block ciphers work
- Understand the AES scheme
- Understand the modes of operation of block ciphers
- Understand padding standards

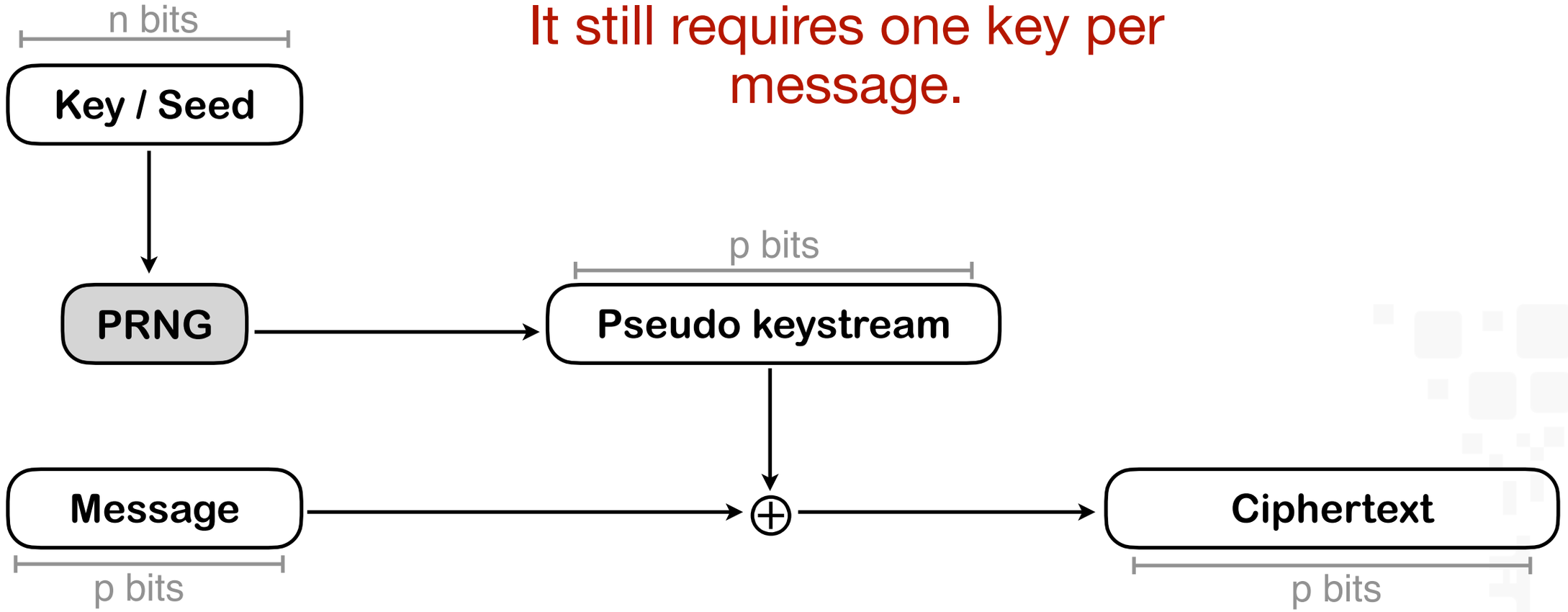
Recap

“Pseudo” One-Time Pad

✓ Key as long as the message

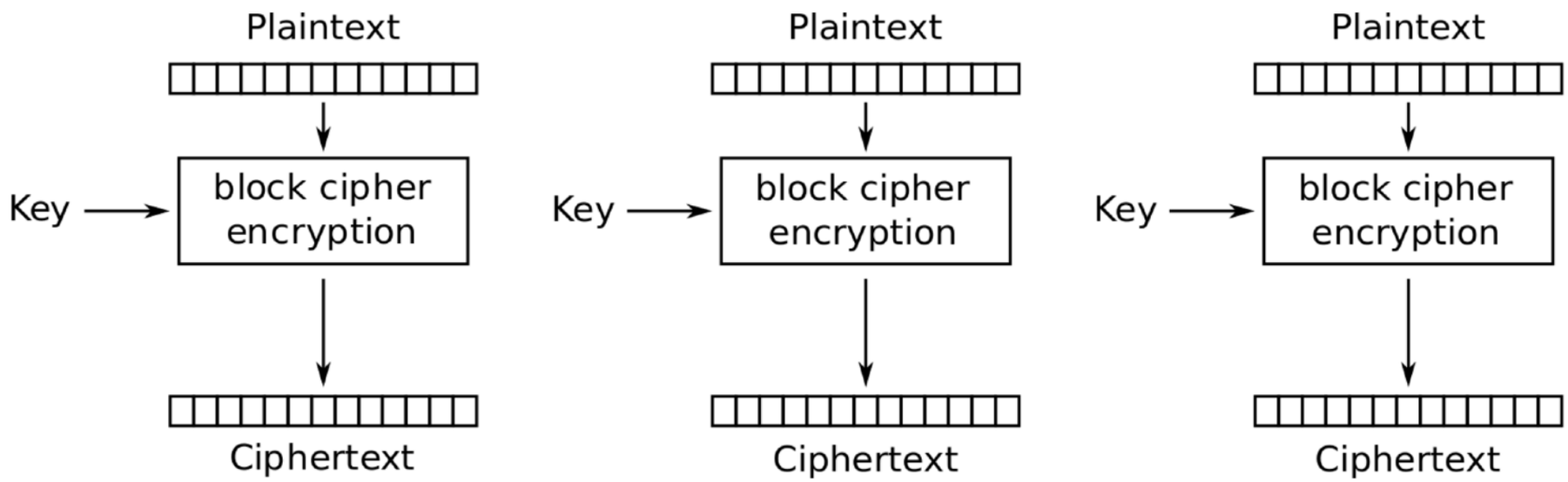
✗ Only secure if each key is used to encrypt once

It still requires one key per message.



Block ciphers

The cryptographic algorithm operates on fixed-length blocks of the input (*e.g.,* plaintext) with the resulting output block (*e.g.,* ciphertext) of the same size.



- Typical block size: 64, 128 and 256 bits
- The plaintext is divided into blocks of equal size
- It is often required to make the plaintext a multiple of the block length - this is done by padding the last block
  - Bit padding
  - Byte padding
  - PKCS#5 and PKCS#7
  - ANSI X9.23

Block ciphers

Diffusion & Confusion

Introduced by Claude Shannon to capture the two basic building blocks for any cryptographic system.

In a strongly ideal cipher, all statistics of the ciphertext are independent of the particular key used.

Confusion: each bit of the cryptogram must be a complex function of the bits of the cleartext.

In this way, it will be difficult to discover unknown parts of the plaintext, even knowing some pieces, and it will also be difficult to deduce parts or all of the key used.

Block ciphers

Diffusion & Confusion

Introduced by Claude Shannon to capture the two basic building blocks for any cryptographic system.

In a strongly ideal cipher, all statistics of the ciphertext are independent of the particular key used.

Diffusion: each bit of the cleartext must affect the highest number of bits in the corresponding ciphertext.

In this way, any small change in the cleartext leads to major changes in the ciphertext, making it difficult to establish relationships between similar original texts and similar cryptograms.

The statistical structure of the plaintext is dissipated into long-range statistic of the ciphertext.

Block ciphers

Feistel Chiper Structure

A practical application of Shannon's principle to develop a product cipher that alternates *confusion* and *diffusion* functions. In particular, it alternates *substitutions* and *permutations*.

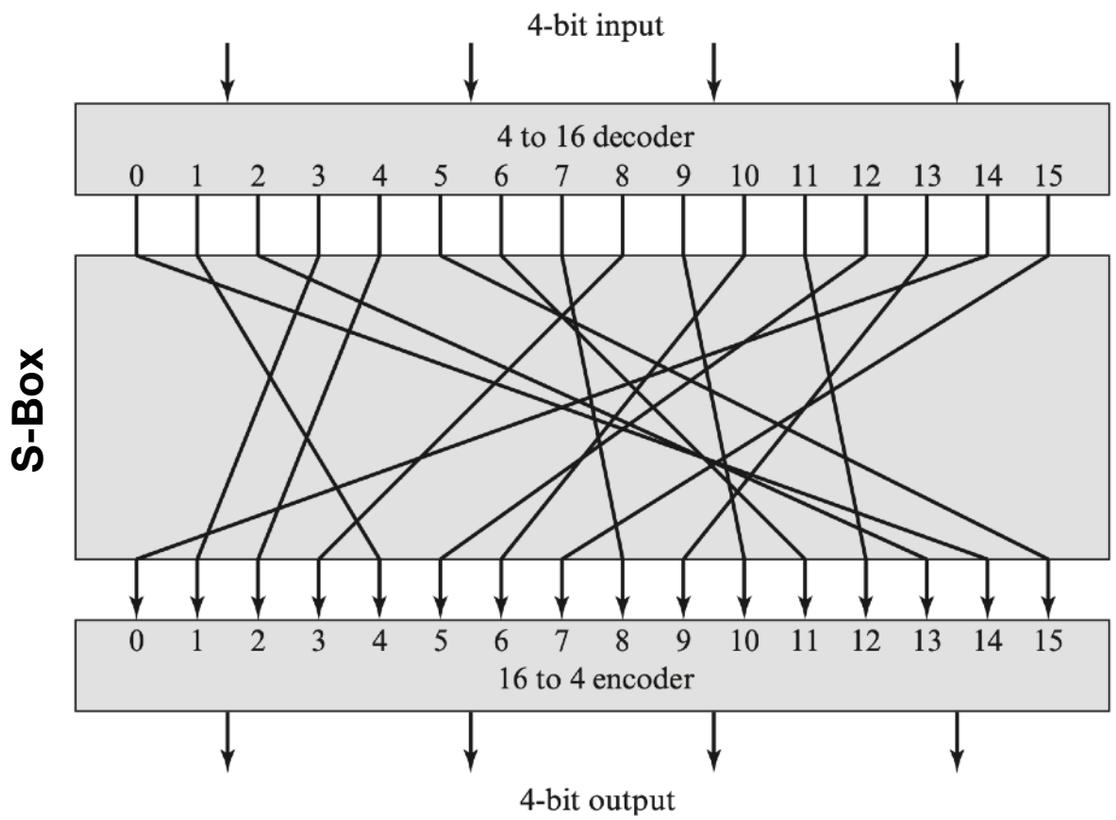
Substitution: Each plaintext element or group of elements is uniquely replaced by a corresponding ciphertext element or group of elements.

Permutation: A sequence of plaintext elements is replaced by a permutation of that sequence. No elements are added, deleted or replaced in the sequence.

Block ciphers

Substitution & Permutation

Substitution



- When operating on binary words, it can be seen as a permutation (exchange of wires) between a decoder/encoder pair
- The mapping itself constitutes the key

Example

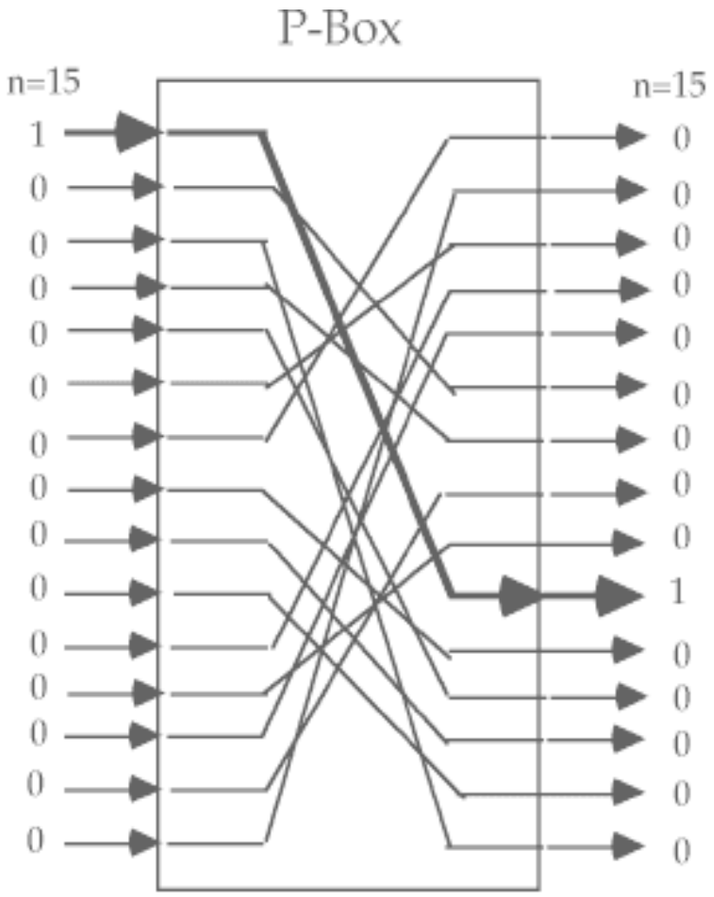
Plaintext	Ciphertext
0000	1110
0001	0100
0010	1101
0011	0001

Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Block ciphers

Substitution & Permutation

Permutation



- The elements in the output are the same as in the input
- Its implementation is quite simple in hardware
  - Not so much in software
- Transpositions are simply an “exchange of wires” between the input and output bits

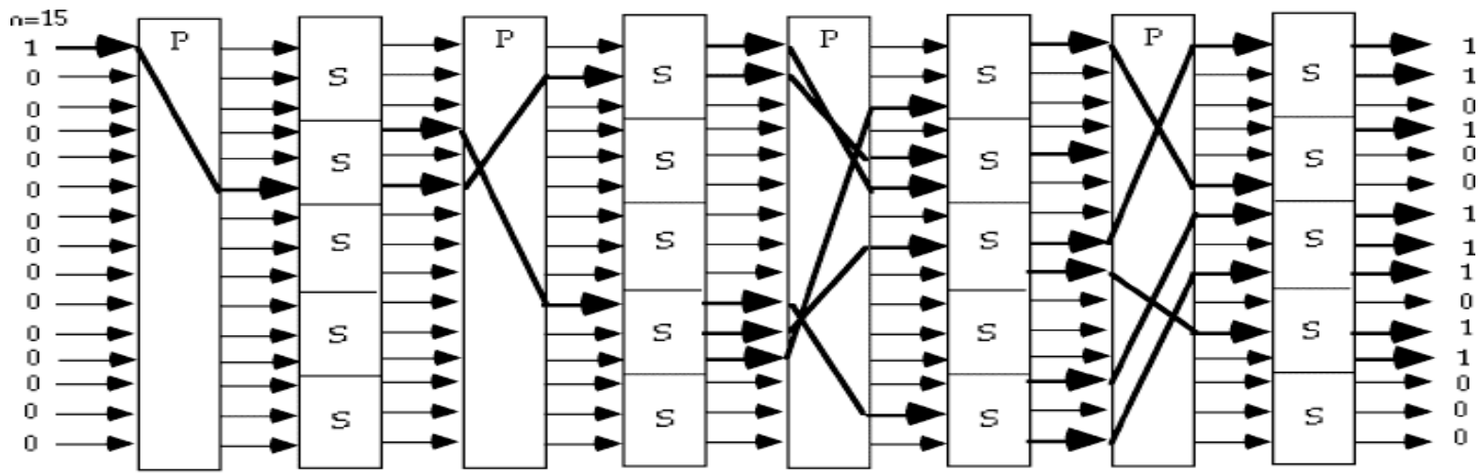
Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Block ciphers

Substitution & Permutation

S-P Networks

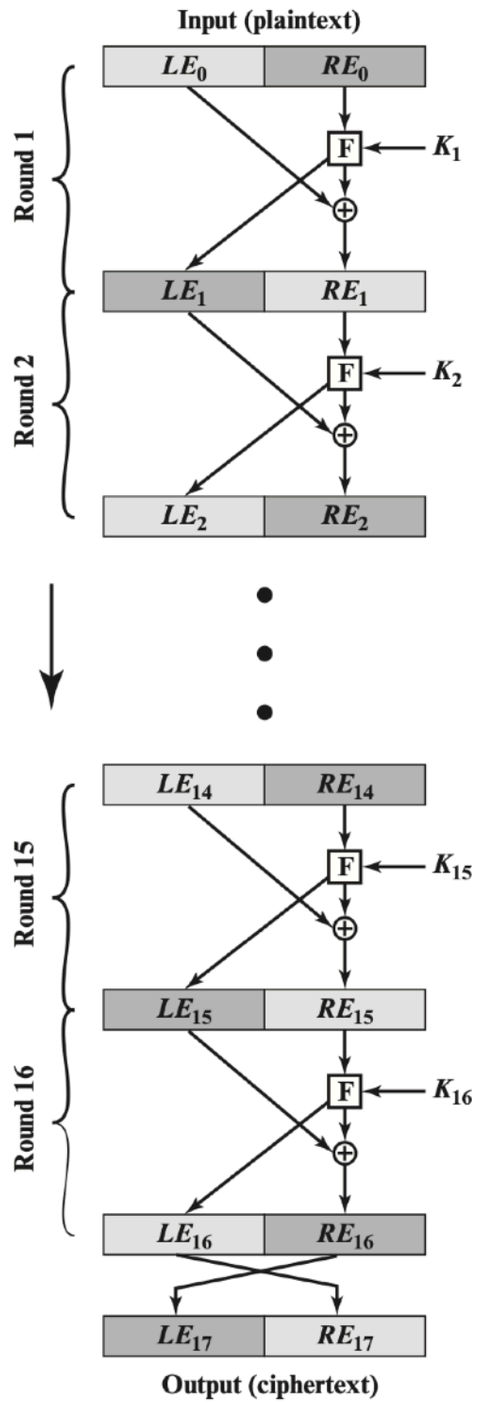
- Substitutions and permutations alone are not enough to build secure ciphers.
  - They are both idempotent operations
- However, by combining both techniques, it is possible to construct a non-idempotent *product cipher* called *round*
- One round alone is not secure enough. But when iterated, the desired security level can be achieved



It is impractical since it requires duplicated devices or programs for decryption.

Block ciphers

Feistel Chiper Structure



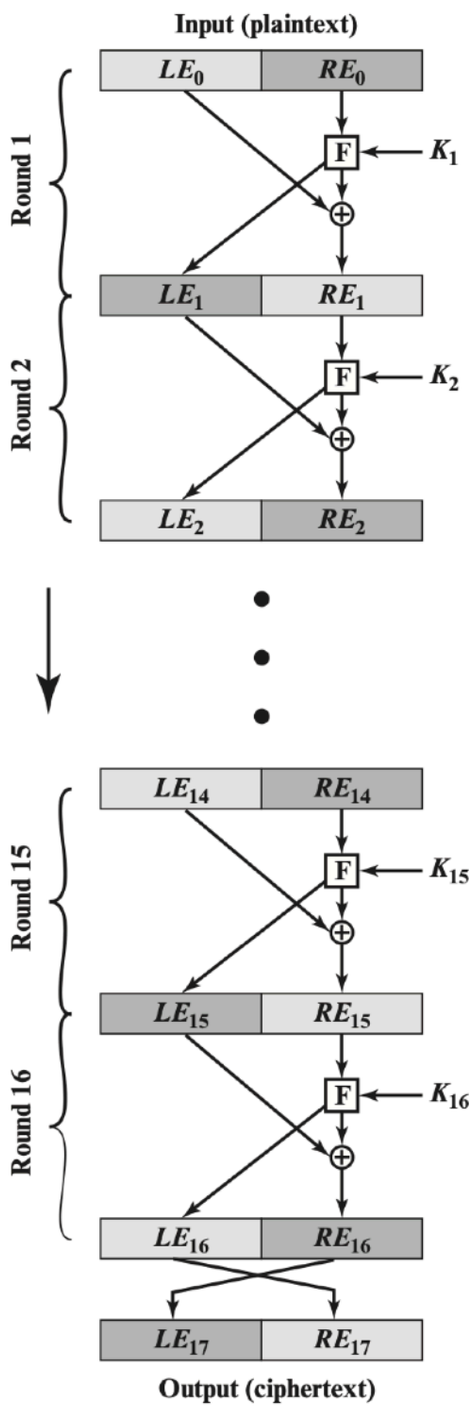
It works in rounds with a substitution and a permutation in each round.

- Algorithm inputs: a plaintext block of length $2w$ bits and a key k
- The plaintext is divided into two halves, LE_0 and RE_0
- The two halves pass through n rounds of processing to produce the ciphertext
- The inputs for each round i are:
  - LE_{i-1} and RE_{i-1} derived from the previous round
  - k_i derived from k, with $k_i \neq k$
- All rounds have the same structure
  - A substitution in the left half by applying the function F to the right half with the subkey k_i and the logical operation XOR between the output of F and the left half
  - A permutation consisting of swapping the two halves

Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Block ciphers

Feistel Chiper Structure



The exact realization of a Feistel network depends on the choice of the following parameters and design features:

Block size: Larger block sizes mean greater security but reduced encryption/decryption speed.

Key size: A larger key size means greater security but may decrease encryption/decryption speed.

Number of rounds: A single round offers inadequate security but multiple rounds offer increasing security.

Subkey generation algorithm: Greater complexity in this algorithm leads to greater difficulty of cryptanalysis.

Round function `F`: Greater complexity generally means greater resistance to cryptanalysis.

Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Block ciphers

Advanced Encryption Standard (AES)

- Announced by the NIST as US FIPS PUB 197 (FIPS 197) in 2001
  - Intended to replace DES and 3DES with an algorithm that is more secure and efficient
    - Proposed by ***Rij****men*-***Dae****men* (*Rijndael*)
- Its design has a strong mathematical foundation on *Finite Fields / Galois Field*
- All operations can be executed by `XOR` and `lookup` tables
  - Efficient for software and hardware implementation

Block ciphers

Advanced Encryption Standard (AES)

- The input to the encryption and decryption algorithms is a single 128-bit block.
  - The block is depicted as a square matrix (`4 x 4`) of bytes filled by columns.
  - The block is copied into the `State` array, which is modified at each stage of encryption of decryptions.

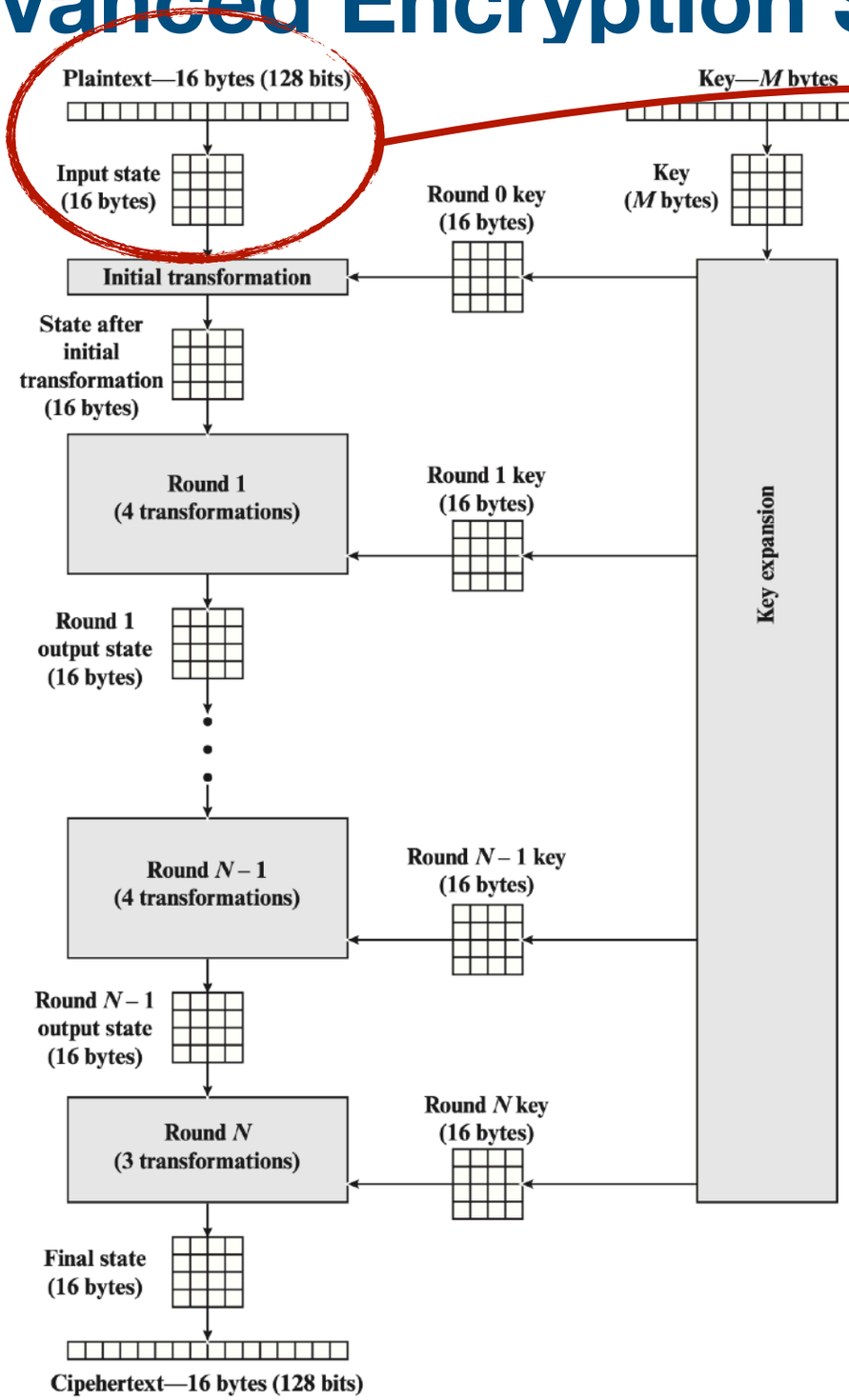
EA	04	65	85
83	45	5D	96
5C	33	98	B0
F0	2D	AD	C5

  - After the final stage, `State` is copied to an output matrix

- The key length can be 128, 192, or 256 bits
  - The key is also depicted as a square matrix of bytes and then expanded into an array of key schedule words
  - Each word is 4 bytes and the total key schedule is 44 words for the 128-bit key

Block ciphers

Advanced Encryption Standard (AES)



Mapping the plaintext block into `State` by columns.

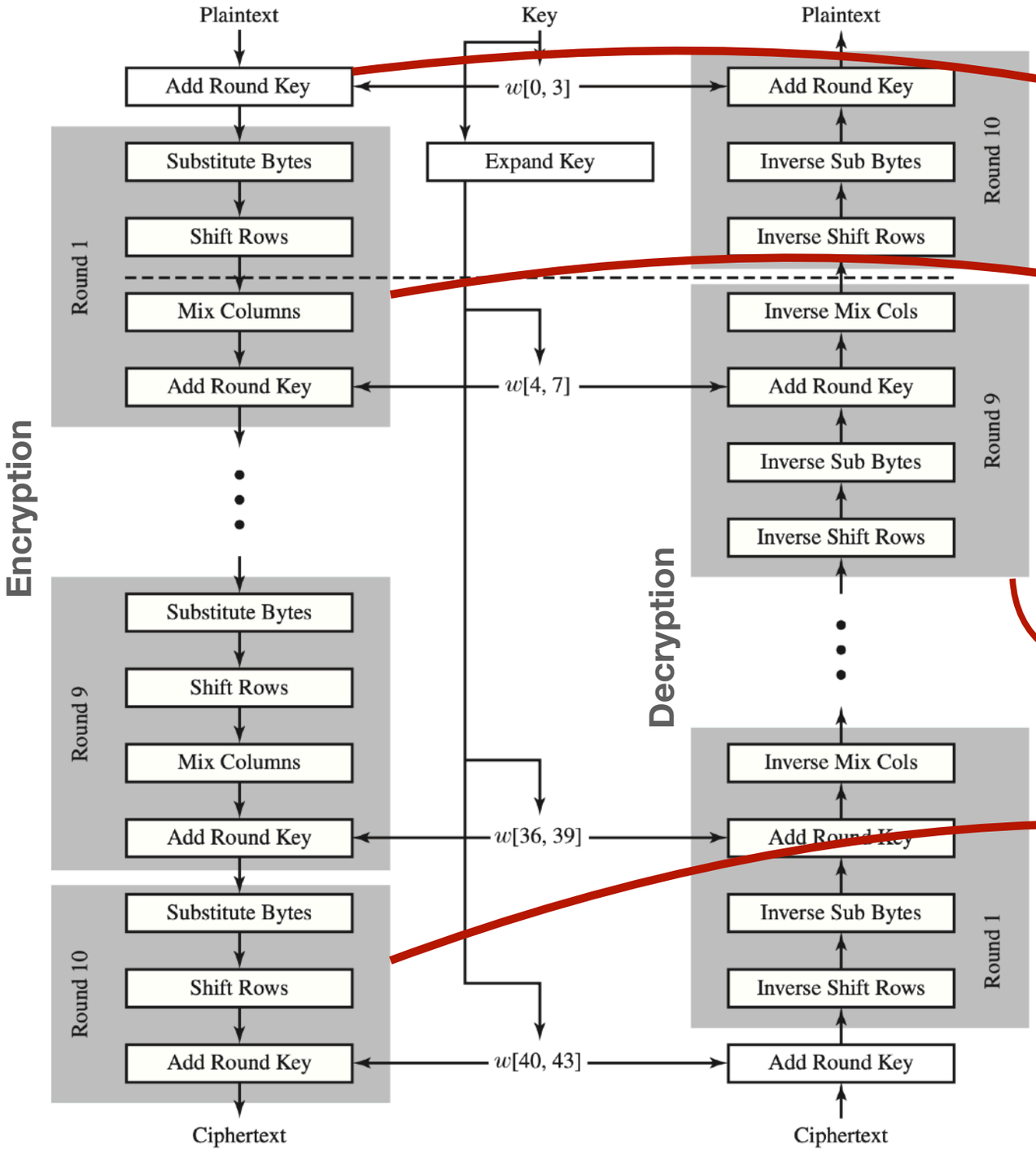
Key length (bytes)	Number of rounds
16	10
24	12
32	14

- It does not use the Feistel structure
  - All block is processed in parallel during each round
- Each round consists of four different stages
  - One permutation
  - Tree Substitutions

Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Block ciphers

Advanced Encryption Standard (AES)



It begins with an `Add Round Key` stage

Followed by nine rounds with four stages:

- `Substitute Bytes`
- `Shift Rows`
- `Mix Columns`
- `Add Round Key`

Each stage is easily reversible

The last round has three stages:

- `Substitute Bytes`
- `Shift Rows`
- `Add Round Key`

Source: Stallings, W. & Brown, L. Computer Security: Principles and Practice

Block ciphers

Advanced Encryption Standard (AES)

Substitute Bytes transformation

Uses an S-Box to perform byte-by-byte substitution of the block. It is done by a simple table lookup in a 16x16 matrix of byte values.

Example

EA	04	65	85
83	45	5D	96
5C	33	98	B0
F0	2D	AD	C5

→

87	F2	4D	97
EC	6E	4C	90
4A	C3	46	E7
8C	D8	95	A6

S-Box

x \ y	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	BI	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

Inverse S-Box

x \ y	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	52	09	6A	D5	30	36	A5	38	BF	40	A3	9E	81	F3	D7	FB
1	7C	E3	39	82	9B	2F	FF	87	34	8E	43	44	C4	DE	E9	CB
2	54	7B	94	32	A6	C2	23	3D	EE	4C	95	0B	42	FA	C3	4E
3	08	2E	A1	66	28	D9	24	B2	76	5B	A2	49	6D	8B	D1	25
4	72	F8	F6	64	86	68	98	16	D4	A4	5C	CC	5D	65	B6	92
5	6C	70	48	50	FD	ED	B9	DA	5E	15	46	57	A7	8D	9D	84
6	90	D8	AB	00	8C	BC	D3	0A	F7	E4	58	05	B8	B3	45	06
7	D0	2C	1E	8F	CA	3F	0F	02	C1	AF	BD	03	01	13	8A	6B
8	3A	91	11	41	4F	67	DC	EA	97	F2	CF	CE	F0	B4	E6	73
9	96	AC	74	22	E7	AD	35	85	E2	F9	37	E8	1C	75	DF	6E
A	47	F1	1A	71	1D	29	C5	89	6F	B7	62	0E	AA	18	BE	1B
B	FC	56	3E	4B	C6	D2	79	20	9A	DB	C0	FE	78	CD	5A	FA
C	1F	DD	A8	33	88	07	C7	31	B1	12	10	59	27	80	EC	5F
D	60	51	7F	A9	19	B5	4A	0D	2D	E5	7A	9F	93	C9	9C	EF
E	A0	E0	3B	4D	AE	2A	F5	B0	C8	EB	BB	3C	83	53	99	61
F	17	2B	04	7E	BA	77	D6	26	E1	69	14	63	55	21	0C	7D

Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

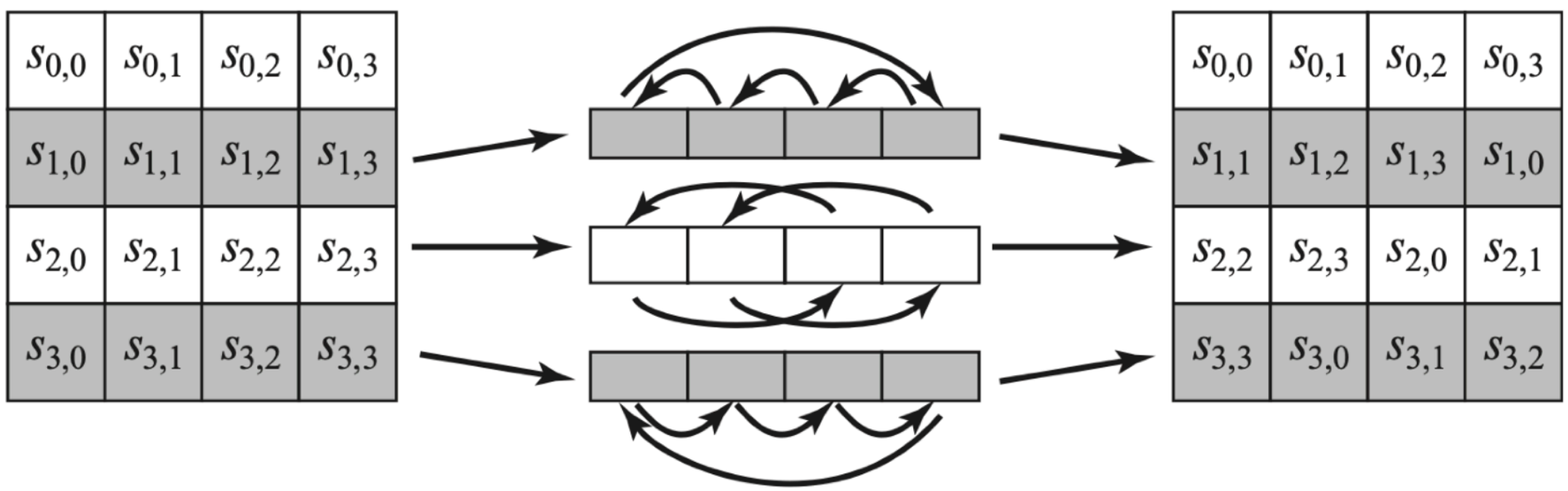
Block ciphers

Advanced Encryption Standard (AES)

Shift Rows transformation

A simple permutation that is performed row by row.

- The first row of State is not altered
- For the second row, a 1-byte circular left shift is performed
- For the third row, a 2-byte circular left shift is performed
- For the fourth row, a 3-byte circular left shift is performed



Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Block ciphers

Advanced Encryption Standard (AES)

Mix Column transformation

A substitution that alters each byte in a column as a function of all of the bytes in the column.

- The transformation can be defined by the following matrix multiplication on `State`.

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} s_{0,0} & s_{0,1} & s_{0,2} & s_{0,3} \\ s_{1,0} & s_{1,1} & s_{1,2} & s_{1,3} \\ s_{2,0} & s_{2,1} & s_{2,2} & s_{2,3} \\ s_{3,0} & s_{3,1} & s_{3,2} & s_{3,3} \end{bmatrix} = \begin{bmatrix} s'_{0,0} & s'_{0,1} & s'_{0,2} & s'_{0,3} \\ s'_{1,0} & s'_{1,1} & s'_{1,2} & s'_{1,3} \\ s'_{2,0} & s'_{2,1} & s'_{2,2} & s'_{2,3} \\ s'_{3,0} & s'_{3,1} & s'_{3,2} & s'_{3,3} \end{bmatrix}$$

- Each element in the product matrix is the sum of products of elements of one row and one column.
- The mapping is designed to provide a good mixing among the bytes of each column.

Block ciphers

Advanced Encryption Standard (AES)

Add Round Key transformation

A simple bitwise `XOR` of the current block with a portion of the expanded key. Although simple, it affects every bit of the `State`.

- The operation is viewed as a columnwise operation between the 4 bytes of a `State` column and one word of the round key.

47	40	A3	4C
37	D4	70	9F
94	E4	3A	42
ED	A5	A6	BC

$\oplus$

AC	19	28	57
77	FA	D1	5C
66	DC	29	00
F3	21	41	6A

$=$

EB	59	8B	1B
40	2E	A1	C3
F2	38	13	42
1E	84	E7	D6

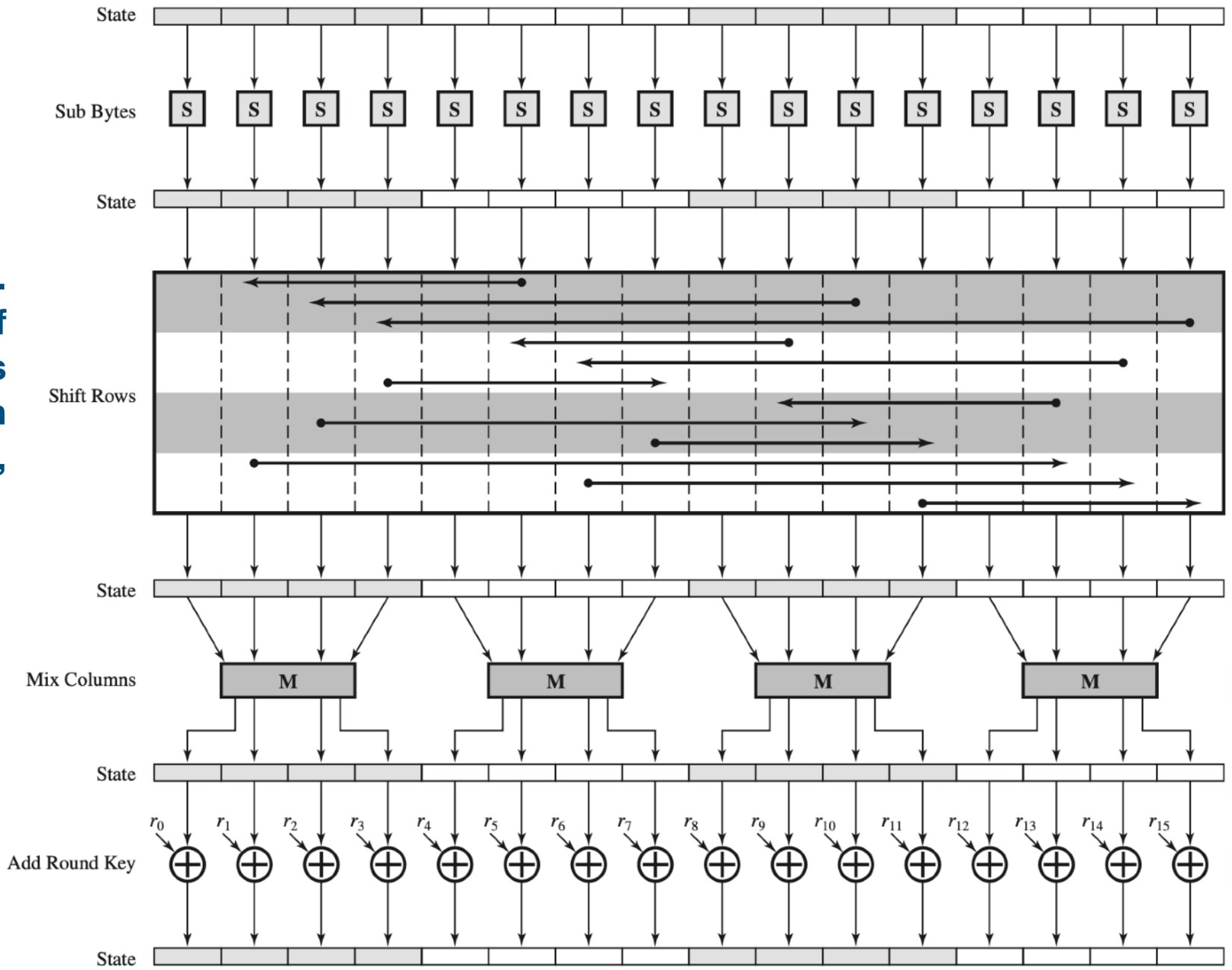
- The first matrix is the `State`, and the second matrix is the round key.
- The `inverse add round key` transformation is identical, as the `XOR` is its own inverse

Block ciphers

Advanced Encryption Standard (AES)

AES Encryption Round

All stages are easily reversible. Therefore, the complexity of the round key expansion, plus the confusion and diffusion generated by the stages, ensure security.



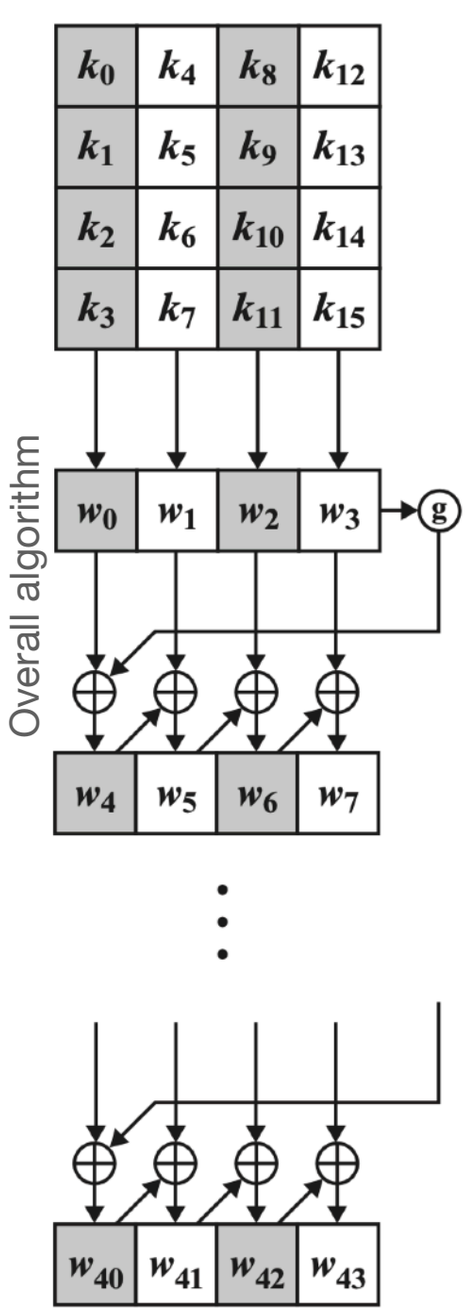
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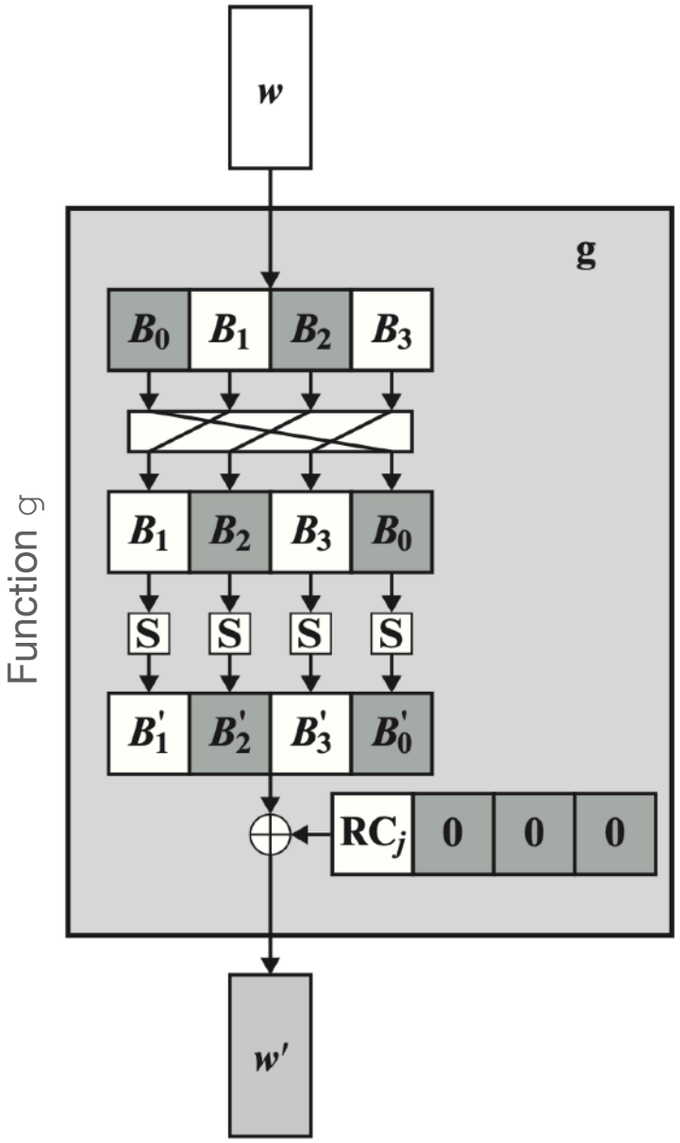
Block ciphers

Advanced Encryption Standard (AES)

AES Key Expansion

- The algorithm takes as input a four-word (16-byte) key and produces a linear array of 44 words (176 bytes)
  - They are used in the initial `Add Round Key` and each of the 10 rounds of the cipher
- The key is copied into the first four words of the expanded key
- The remainder of the expanded key is filled in four words at a time





Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Block ciphers

Modes of operation

To apply a block cipher in various application types, five operation modes have been defined by NIST.

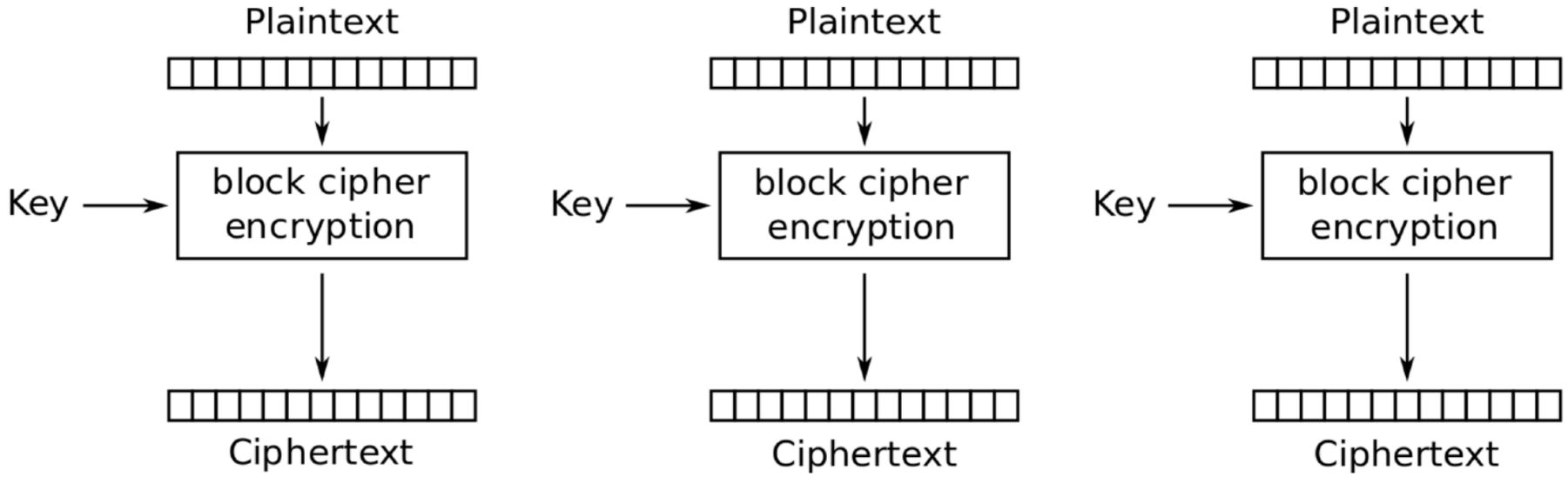
- **Electronic Codebook (ECB)**
- **Cipher Block Chaining (CBC)**
- **Cipher Feedback (CFB)**
- **Output Feedback (OFB)**
- **Counter (CTR)**

Block ciphers

Modes of operation

Electronic Codebook (ECB)

Each block of the plaintext bits is encoded independently using the same key.



Typical application

- Transmission of single values.

Block ciphers

Modes of operation

Electronic Codebook (ECB)

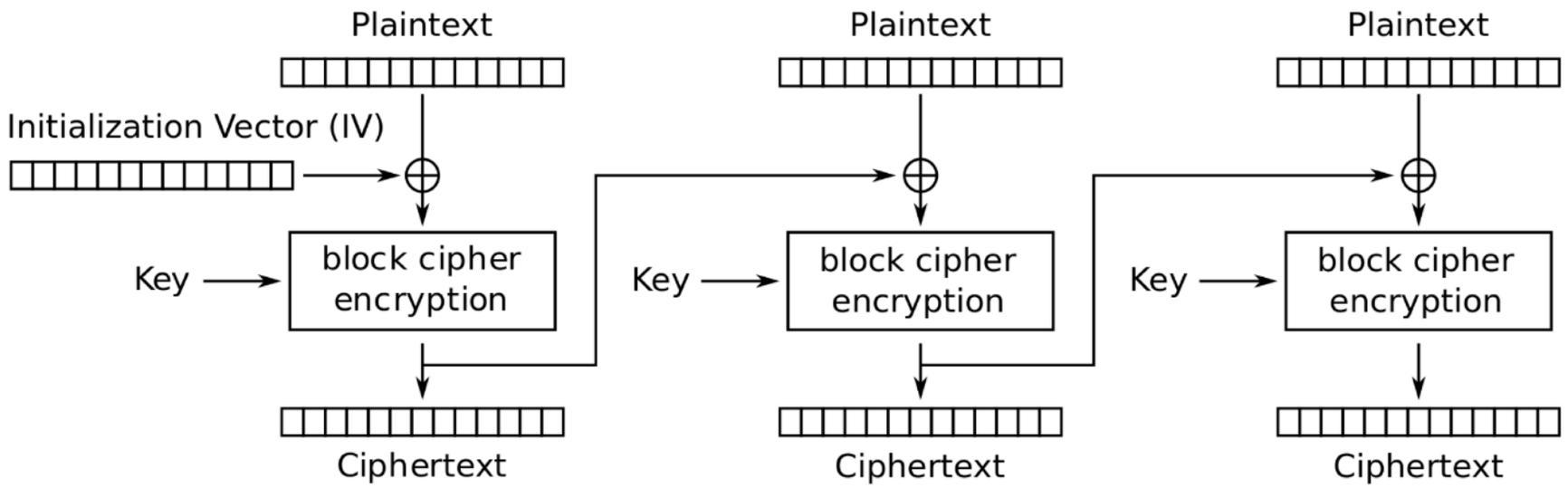
- Repeated blocks are detected (*code book attack*)
  - Two identical blocks in the cleartext produce two identical blocks in the ciphertext
- Vulnerable to replay and substitution attacks
- A one-bit error in one block of the cryptogram affects a single block of the plaintext after decryption
- It should only be used to encrypt messages in a single block (or a few)

Block ciphers

Modes of operation

Cipher Block Chaining (CBC)

The input to the encryption algorithm is the XOR of the next block of plaintext and the preceding block of ciphertext.



Typical application

- General-purpose block-oriented transmission
- Authentication

Block ciphers

Modes of operation

Cipher Block Chaining (CBC)

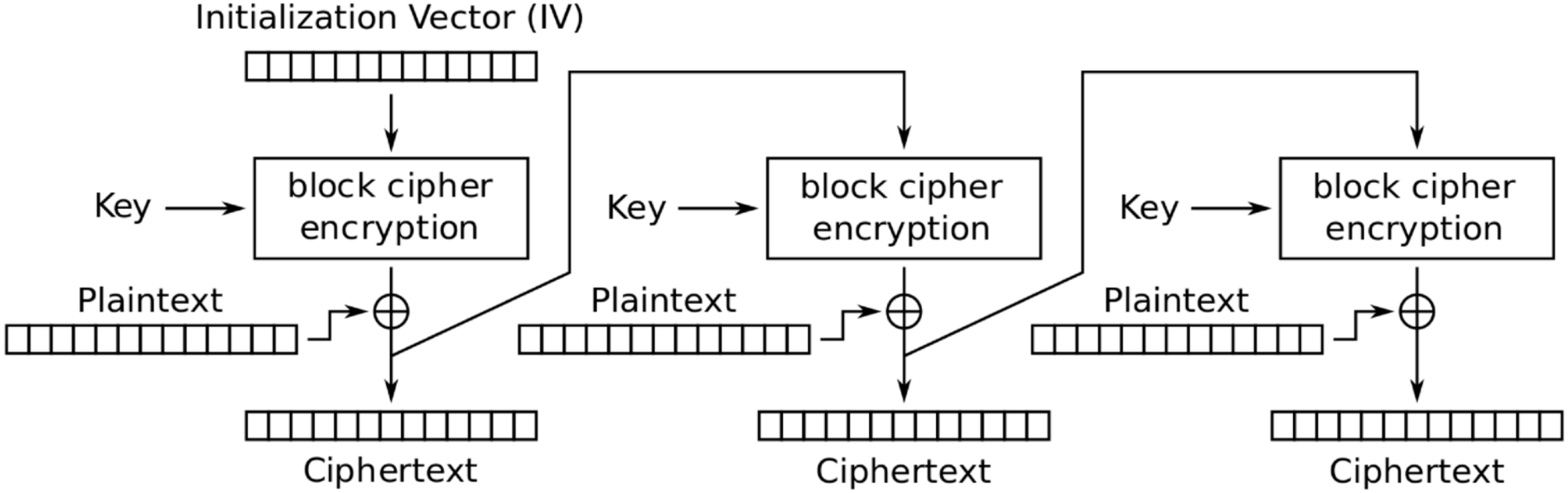
- Its main objective is to prevent the repetition patterns of ECB
- The value of the Initialization Vector (IV) used to process the first block can be secret or not
  - If the IV is sent in cleartext, an attacker can change bits of the first block by changing the respective bits of the IV
  - If the IV is fixed, the ciphertext of two messages with a common prefix will preserve this property
- An error in one block of the ciphertext corrupts two blocks after decryption
- The chaining process means that encrypting a block depends on all the blocks that precede it
  - It can be used in message authentication

Block ciphers

Modes of operation

Cipher Feedback (CFB)

The input is processed n bits at a time. Preceding ciphertext is used as input to the encryption algorithm to produce pseudorandom output, which is XORed with the plaintext to produce the next unit of ciphertext.



Typical application

- General-purpose stream-oriented transmission
- Authentication

Block ciphers

Modes of operation

Cipher Feedback (CFB)

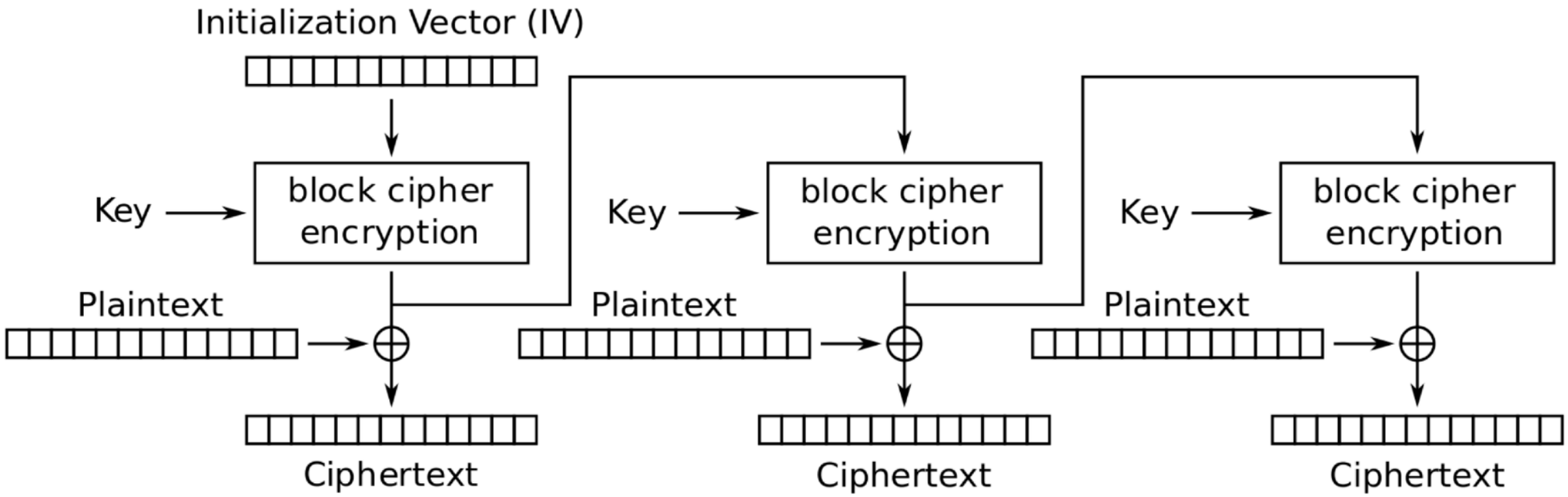
- A mode that transforms a block cipher into a stream cipher
- The number of bits in the feedback is variable (*i.e.*, n=1, 8 or 64)
  - The feedback transfers the n most significant bits to the least significant bits (with a shift of the rest)
- The IV must be unique for each use and can be sent in cleartext
- The key sequence depends on the IV, the cipher key, and all the cleartext already encrypted
- An error in one bit of the ciphertext affects the corresponding bit in the block and all bits in the next block

Block ciphers

Modes of operation

Output Feedback (OFB)

Similar to CFB, except that the input to the encryption algorithm is the preceding encryption output, and full blocks are used.



Typical application

- Stream-oriented transmission over noisy channels, e.g., satellite communications

Block ciphers

Modes of operation

Output Feedback (OFB)

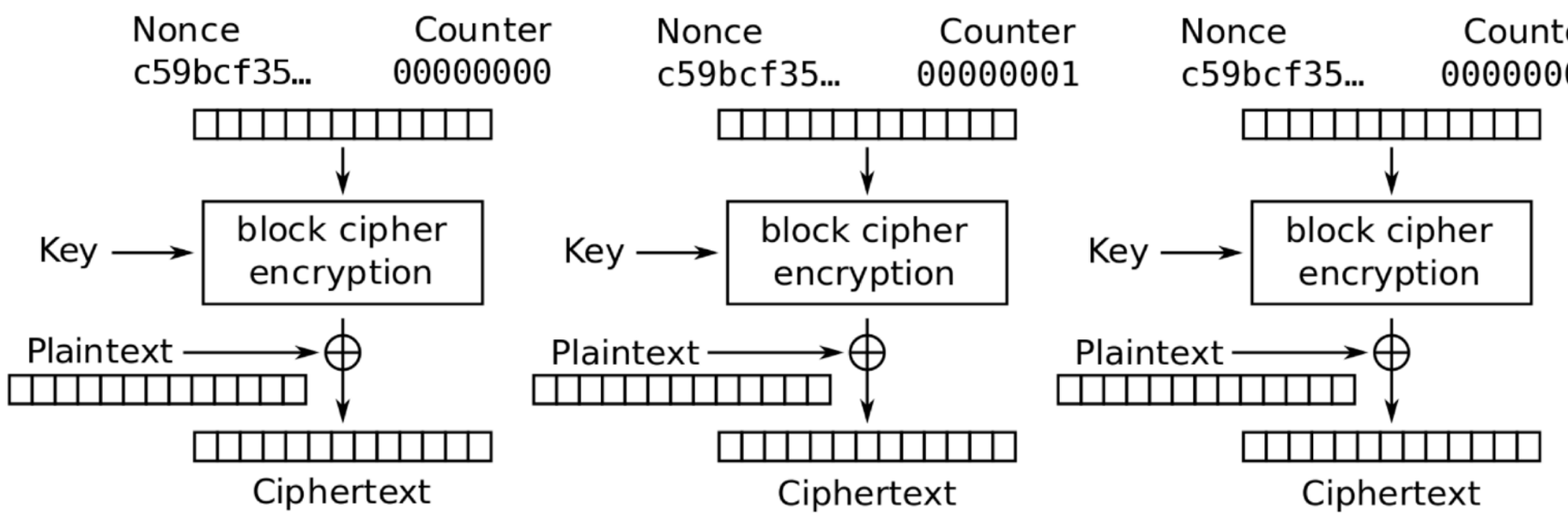
- A mode that transforms a block cipher into a stream cipher
- The key sequence is obtained by iterating the cipher over an initial block, *i.e.*, the IV
- The key sequence is independent of the message, so it can be processed regardless of the message availability
- Bit errors in the ciphertext only affect the respective bits in the decrypted plaintext
  - An advantage over CFB, since bit errors do not propagate
  - However, it is more vulnerable to message stream modification attack

Block ciphers

Modes of operation

Counter (CTR)

Each block of plaintext is XORed with an encrypted counter. The counter is incremented for each subsequent block.



Typical application

- General-purpose block-oriented transmission
- Useful for high-speed requirements

Block ciphers

Modes of operation

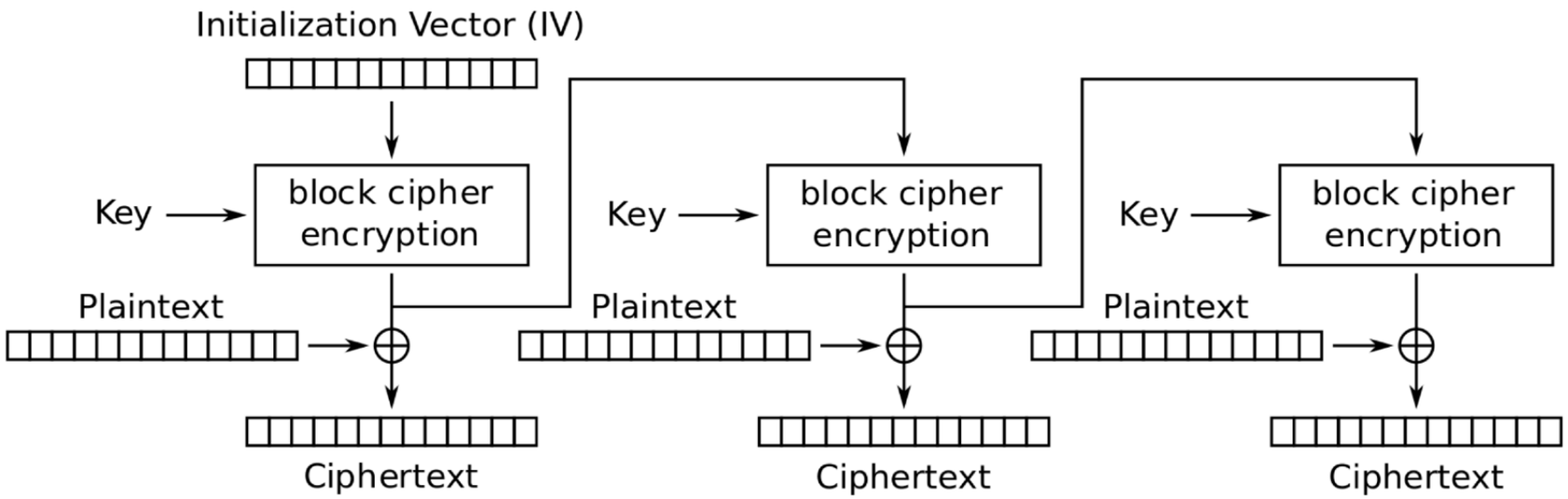
Counter (CTR)

- Efficient hardware and software implementations
- Simple and secure
- The *Nonce* (IV) and *Counter* can be combined in different ways (concatenated, XORed, etc.)
- The requirement of the *Counter* is to produce different values for all the blocks
  - A counter is the simplest way to do it
- There is no dependency between processing blocks (they can be processed in parallel, randomly, etc.)

Block ciphers

Initialization Vector

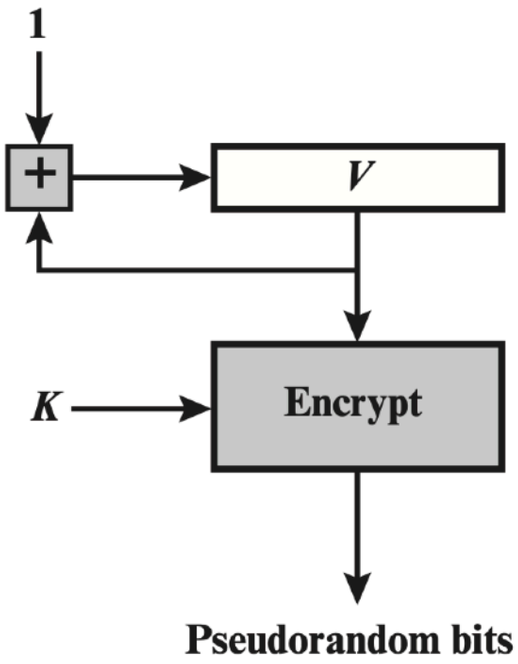
It is important to note that the Initialization Vector (IV) is not supposed to be a secret. However, it must be unique per encrypted message and unpredictable.



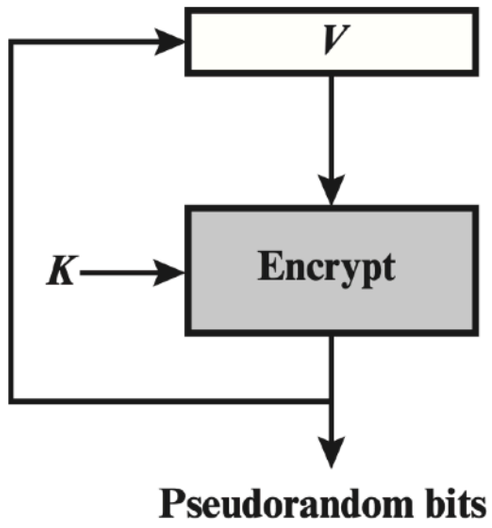
Block ciphers

Pseudorandom Generators with Block Ciphers

CTR mode



OFB mode



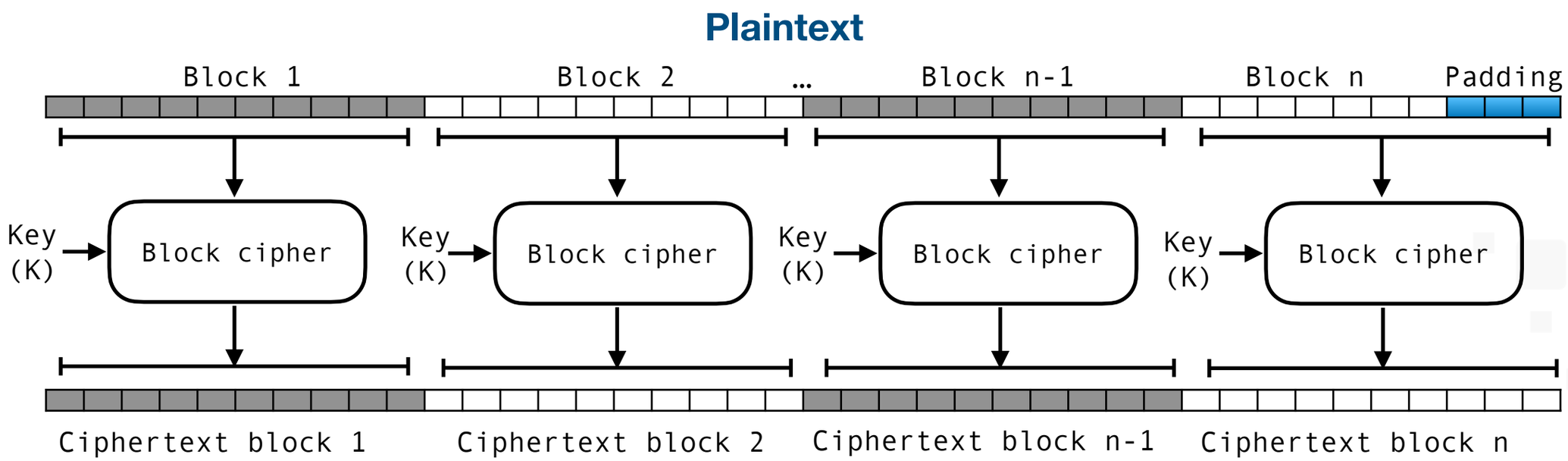
What constitutes the seed?

The seed has two parts: the `encryption key` and a value `V`

Block ciphers

Padding

- Block ciphers transform messages of fixed size into ciphertexts of the same size
- If the message isn't a multiple of the block size, it must be padded
  - This is not an issue for the modes that can be used for stream ciphers



Block ciphers

Padding

- A padding scheme needs to clearly mark where the padding starts
  - So decryption can remove the padded data

Padding schemes

PKCS#7 / PKCS#5: Add `N` bytes with value `N`, where `N` is the number of bytes added

ANSI X9.23: Appends `N-1` bytes with value `0` and the last byte with the value `N`, where `N` is the number of padded bytes

Bit padding: Add a bit `1` followed by enough bits `0` to reach the required length (used in some hash functions)

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Data Security

Master's in Cybersecurity

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